MIL-G-16408B(SHIPS)
27 August 1953
SUPERSEDING
MIL-G-16408A
18 June 1952

MILITARY SPECIFICATION

GAGE, LIQUID LEVEL, TAPE AND FLOAT, DIRECT READING

(FOR SHIPBOARD USE)

1. SCOPE

1.1 This specification covers float actuated, direct reading marine tank gages suitable for cargo fuel tanks.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 - Boxes; Wood-Wirebound, for Domestic Shipment.
QQ-P-416 - Plating; Cadmium (Electro-Deposited).
QQ-S-781 - Strapping, Flat, Steel.
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment.

MILITARY

JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed,
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type).
MIL-P-116 - Preservation, Methods of.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-139 - Packaging and Packing for Overseas Shipment - Plywood, Container-Grade.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.
JAN-P-658 - Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electro-Mechanical).
MIL-N-894 - Nickel-Copper-Alloy: Bars, Forgings, Plates, Rods, Shapes, Sheets, Strips, and Wire.
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-B-994 - Brass, Naval, Wrought.
JAN-S-1160 - Steel, Spring, Round or Square (Including Rectangular Sections Having Thicknesses Greater Than Half Their Width).
MIL-I-15024 - Identification Plates, Information Plates,and Marking Information for Identification of Electrical, Electronic, and Mechanical Equipment.
MIL-B-15071 - Books, Instruction: Preparation, Contents, and Approval.
MIL-B-16444 - Bronze, Hydraulic (Ounce Metal) Castings.
MIL-F-17292 - Format for List of Repair (Spare) Parts for Shipboard, Mechanical and Electrical Equipment.
MIL-P-17577 - Packing Material Asbestos, Rod, Braided.
MIL-P-20085 - Packing, Sheet, Plant- or Animal-Fiber.

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NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Nonsparking materials shall be used throughout.

3.1.2 The float hanger, tape sheave, crank, float guide bushings, and any other brass part not hereinafter specifically identified shall be of wrought Naval brass conforming to Specification MIL-B-994.

3.1.3 The gaskets shall be plant- or animal-fiber sheet in accordance with Specification MIL-P-20085.

3.1.4 Packing shall be braided asbestos in accordance with type I of Specification MIL-P-17577.

3.1.5 The body of the head, including the access cover, shall be cast hydraulic bronze conforming to Specification MIL-B-16444.

3.1.6 The graduated tape and float guide wires shall be of nickel-copper alloy conforming to Specification MIL-N-894.

3.1.7 Guide wire tension springs shall be of spring steel conforming to Specification JAN-S-1160, and cadmium plated in accordance with Specification QQ-P-416.

3.1.8 The float shall be of a gasoline resistant plastic, and the counterweight shall be of lead.

3.1.9 The foot piece and counterweight access plug piping shall be of galvanized steel.

3.2 Construction.- Each gage shall consist of a gage head, float, float guide wires, counterweight, graduated tape, and foot piece, including access plug piping.

3.2.1 The gage head shall consist of the head body, an access cover, float hanger, float guide wire tension springs, and a cranksheave assembly. A gasket as specified in 3.1.3 shall be provided to permit separation of the gage head body from the deck.

3.2.2 The gage head access cover shall be hinged, and shall contain a reading window provided with a wiper for keeping the inside surface of the window clean.

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3.2.3 A grooved sheave with crank attached shall form a part of the gage head, and shall be capable of manually raising and lowering the gage float. A float hanger or locking device shall be provided for locking the gage float in the uppermost position when the gage is not in use.

3.2.4 Two float guide wires shall be provided for each gage. A spring assembly shall be provided for keeping the float guide wires taut.

3.2.5 The float shall be of sufficient weight to overcome the weight of the couterweight when the float is lowered to the level of the liquid in the tank.

3.2.6 Unless otherwise specified in the contract or order, the tape shall be 60 feet in length, and shall be graduated in feet and inches for a minimum length of 60 feet.

3.3 Designation and marking. -

3.3.1 On each gage shall be a identification plate in accordance with type A or B of Specification MIL-I-15024. Plastic identification plates are not acceptable. Plates shall be attached to a part of the equipment which will not ordinarily be renewed during its normal service life, and shall be located in a readily accessible position where easily read.

3.3.2 Data on the identification plates shall include the following items:

(a) Manufacturer's name, model number, serial number, and contract number.
(b) Stock number.
(c) Blank space for official stamping by the Government inspector.

(If a stock number does not appear in the schedule, it will be furnished to the contractor by the bureau or agency concerned upon written request, with which request there shall be furnished the contractor's drawing for the article concerned.)

3.4 Drawings. - Drawings shall be furnished in accordance with type I of Specification MIL-D-963.

3.5 Instruction books. - Instruction books shall be furnished in accordance with type D of Specification MIL-B-15071. Instruction books shall completely describe the operation of the unit and cover installation procedures, instructions for adjustments, and maintenance.

3.6 Special tools. - If required, special tools or equipment shall be furnished for adjustment, disassembling and servicing of equipment.

3.7 Repair parts. - When repair parts are required, they shall be as specified (see 6.1), and satisfactory to the bureau or agency concerned.

3.8 Workmanship. - Gages shall be free of imperfections which may impair serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance. -

4.1.1 Lot. - For purposes of sampling, a lot shall consist of all gages of the same design offered for delivery at the same time.

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4.1.2 Sampling for inspection.- A random sample of gages shall be selected in accordance with table I from each lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Standard MIL-STD-105.

Table I - Sampling for visual and dimensional inspection
AQL (approx.) = 2.5 percent defective.

Number of gages in lot	Number of gages in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 to 500	35	2	3
501 and over	50	3	4

4.2 Inspection at the place of manufacture.- Each sample gage selected in accordance with 4.1.2 shall be inspected by the Government inspector to determine conformance with this specification and approved drawings as to material, workmanship, and dimensions. Any gage in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. A rejected lot may be resubmitted for Government inspection after the manufacturer has reinspected the entire lot and removed all nonconforming gages.

4.3 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment - immediate use.- Tank gages shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.1.2 For domestic shipment and storage or overseas shipment.- Tank gages shall be preserved and packaged in accordance with method I of Specification MIL-P-116. Preservative type P-8 shall be coated on all corrodible metal surfaces. Detached components removed to facilitate shipment shall be packaged in accordance with method I or III as applicable.

5.2 Packing.-

5.2.1 For domestic shipment - immediate use.- Tank gages shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall conform to the Consolidated Freight Classification Rules or rules of other carriers as applicable to the mode of transportation.

5.2.2 For domestic shipment and storage.- Gages, packaged as specified in 5.1.2, shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated or solid fiber boxes conforming to Specifications NN-B-591, NN-B-601, NN-B-621, NN-B-631, LLL-B-631 and LLL-B-636, respectively. Fiberboard shall conform to the special requirements of the applicable specification. The gross weight of wood and wood-cleated boxes shall not exceed 200 pounds.

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5.2.3 For overseas shipment.- Gages, packaged as specified in 5.1.2, shall be packed in cleated plywood, nailed wood or wirebound boxes conforming to Specifications JAN-P-105, JAN-P-106, and MIL-B-107, respectively. Plywood shall conform to type A or B, condition I of Specification JAN-P-139. Boxes shall be lined with a sealed waterproof case liner made of material conforming to Specification JAN-P-125. Liners shall be sealed with an adhesive conforming to Specification MIL-A-140. Seams and closures shall have strength and water resistance equal to that of the body material. Shipping containers shall be closed and strapped in accordance with the applicable container specification. Flat steel strapping shall conform to class A or B of Specification QQ-S-781. The gross weight of wood boxes shall not exceed 150 pounds; of fiberboard boxes, 70 pounds.

5.3 Repair parts.- Repair parts and tools (onboard and stock) shall be preserved and packaged in accordance with Specification MIL-P-116, using Specification JAN-P-658 as a guide. Onboard repair parts and stock repair parts shall be packed in shipping containers as specified in 5.2.2 or 5.2.3, as applicable.

5.4 List of repair parts and tools.- An index list of repair parts and tools conforming to Specification MIL-F-17292 shall be inserted in each repair parts shipping container.

5.5 Technical publications.- Instruction books or pamphlets shall be packaged and packed in accordance with Specification MIL-B-15071.

5.6 Marking.- In addition to any special marking required in the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract or order.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Repair parts required (see 3.7).
(c) Whether preparation is required for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment (see 5.1 and 5.2).
(d) That a preliminary drawing of the tank gage, including list of material, shall accompany bids.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Bureau of Ships
Other interest:
None

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